



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN BIOMEDICAL
ENGINEERING

EBM 200 BIOMATERIALS

1ST SEMESTER 2020/2021

TIME: 2 HOURS

INSTRUCTIONS:

Answer Question ONE and Any ^{TWO} ~~THREE~~ Questions

Question 1

- (a) Property: this is a material trait in terms of the kind and magnitude of response to a specific imposed stimulus. All-important properties of solid materials may be grouped into six different categories'. State and define all SIX properties. *M - mechanical, E - electrical, TH - thermal, M - magnetic, O - optical, D - dielectric* (12 marks)
- (b) Define the following terms and include their equations.
- (i) Eutectic $L \rightarrow S_1 + S_2$ (3 marks)
 - (ii) Eutectoid $S_1 \rightarrow S_2 + S_3$ (3 marks)
 - (iii) Peritectic $L + S_1 \rightarrow S_2$ (3 marks)
 - (iv) Fatigue - ~~stress~~ under cyclic stress (2 marks)
 - (v) Creep - deformation under $\dot{\epsilon} = \dot{\epsilon}_0 e^{-Q/RT}$ (2 marks)
- (c) A cylinder rod of brass originally 14mm in diameter is to be cold worked by drawing. The circular section will be maintained during deformation. A cold-worked tensile strength in excess of 400 MPa and a ductility of at least 11%EL are desired. Furthermore, the final diameter must be 7.5mm. Explain how this may be accomplished.

(5 marks)

Question 2

- (a) Give FIVE (5) examples of how materials are dependent of properties. (5 marks)
- (b) Explain the SIX (6) different classes of materials. *Metals, Polymers, Ceramics, Composites, Biomaterials, Semiconductors* (6 marks)
- (c) Describe and outline the different aspects of the iron/steel alloy including its changes when cooled. *Ductility, Tensile strength, Yield strength* (5 marks)
- (d) Discuss the different mechanical properties of materials (include illustrations and equations where appropriate). *Stress-strain, Compression & tension, Young's modulus, ductility $\rightarrow$ % elongation* (4 marks)

INVOLVEMENT IN EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Deformation elastic & plastic

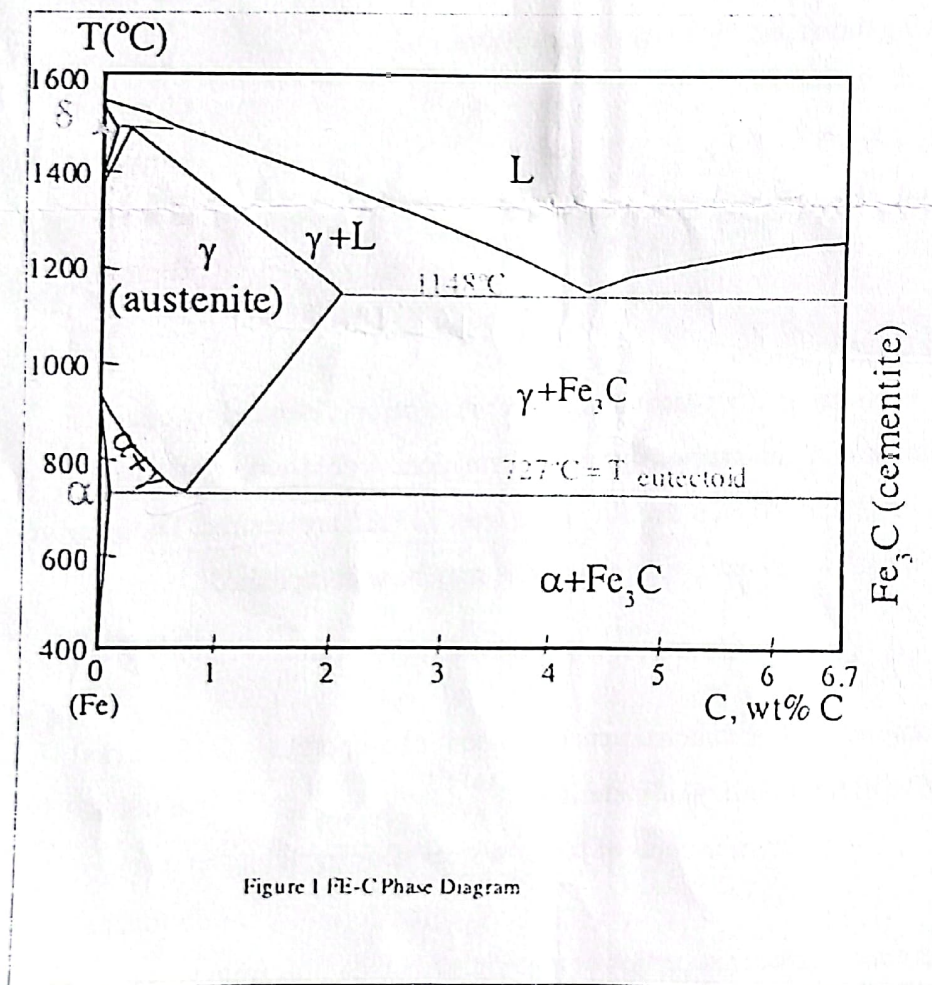
Question 3

a. Define the following terms when describing phase diagrams:

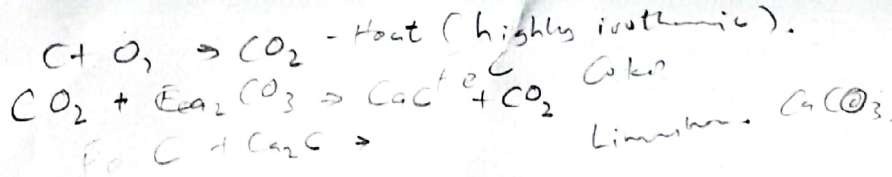
- i. Lever Rule (2 marks)
- ii. Liquidus line (2 marks)
- iii. Hypoeutectoid alloy (3 marks)
- iv. Hypereutectoid alloy (3 marks)

b. For a 99.65 wt% Fe-0.35wt% C alloy. At a temperature just below the eutectoid, Using Figure 1, determine the following:

- i. The compositions of Fe_3C and ferrite (α). (3 marks)
- ii. The amount of cementite (in grams) that forms in 100 g of steel. (3 marks)
- iii. The amounts of pearlite and proeutectoid ferrite (α) in the 100 g. (4 marks)



Question 4



Austenite
 Ferrite
 Cementite
 Pearlite
 Martensite
 Spherulite

(a) Describe how steel is refined from ore. In your explanation define what happens at each stage. Include a diagram in your answer. (5 marks)

(b) Discuss the different types of cast iron. (In your answer include the function of each type, advantages and disadvantages of each. Grey iron, White iron, Malleable iron. (6 marks)

(c) Describe the THREE (3) different methods used for fabrication of metals. (4 marks)

(d) Discuss the different types of annealing. Full annealing (5 marks)

- Process annealing
 - Spheroidization
 - Normalizing
 - Stress-relief

Question 5

Materials science looks at other properties that affect function. Explain each on these properties (include equations where necessary)

(a) Heat Capacity

(b) Ferromagnetic Ceramics

(c) Piezoelectric Materials

(d) Refraction

(e) Reflection of light for metals

(f) Scattering of light in polymers

(g) Semiconductor Laser application

(Polymers have high heat capacity)
 - rapid dipole moment induced.
 - stress induced.
 - high n Optical (3 marks)
 - (2 marks)
 - it passes through (2 marks)
 - (3 marks)
 - some is reflected some absorbed (4 marks)
 - (3 marks)
 - (3 marks)

$$Q = \frac{I}{m \cdot K}$$

$$Q = C \cdot v \cdot d \cdot T$$